



Methodological procedures in Contrastive Linguistics: Correlative considerations

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Abstract

The Saussurian dichotomy 'langue vs. parole uncovers, within the field of contrastive linguistics, the existence of the language level and of the subcorpora level; two levels where placements are realized. With the methodological principles in general and the principles of homogeneity in particular elaborated by Pan and Tham (2007), this study seeks to establish links between items at language level and at subcorpora level. Thanks to the descriptive approach, the five placements identified at each level result in the establishment of seven correlations; some as being simple and others as being complex.

Article info

Received

August 17.2025

Accepted

October 18.2025

Keyword:

- ✓ *Correlations*
- ✓ *Language placements*
- ✓ *Subcorpora placements*

1. Introduction

Research in contrastive linguistics is going beyond the traditional boundaries of general linguistics in the quest for its own identity. The needs for methodology clarification within this field therefore arise stemming partly from its acknowledgement by some scholars as an autonomous discipline and partly from the peculiarities of its thematic scope, objectives, and its epistemological nature. As it is broadly accepted, contrastive linguistics focuses on bringing out similarities and differences between two or more linguistic systems. Therefore, the contrastivist cannot work on all language aspects in a single study as it is rather advised to tackle just one language aspect at a time. This methodological prescription calls for the use corpora that are made up of subcorpora according to Fonkoua and Bayiha (2023).

Scholars mostly consider research in Contrastive Linguistics as corpus-based because of its link to general linguistics, of its subject matters and of its investigation requirements. Much has already been done in this field: several authors designed corpora useful to forthcoming studies [English-Swedish Parallel Corpus (ESPC) accessible in <https://spraakbanken.gu.se/korp/?mode=parallel#?corpus=espc-sv> ; Oslo Multilingual Corpus (OMC) accessible in <https://www.hf.uio.no/ilos/english/services/knownledge-resources/omc> ; English-Norwegian Parallel Corpus (ENPC) accessible in <https://varieng.helsinki.fi/CoRD/corpora/ENPC/background.html> ; Punjabi-English (PunEng) Parallel Corpus in Deep, Kumar, and Goyal (2018); the English – Thai parallel corpus in Lowphansirikul et al. (2022)];

meanwhile other have identified and discussed on corpus types within the discipline [Johansson (1998); Altenberg and Granger (2002); Nadvornikova (2017)]; and other have focused on carrying out practical studies using existing corpora [Johansson (2003); part III in Xiao (2010);]; and other works include the above features [Mikhailov and Cooper (2016)]. Furthermore, it is well established that most previous works assimilate language to the corpus while contrasting. In other words, previous works hardly draw a clear line between languages in comparison and the corpus used for analysis. So, there is a gap that remains void and paths that are still unexplored. With regard to the above, this study seeks to explore different links existing between methodological tools in the field of contrastive studies.

This study intends to present and elaborate on various correlations related to methodological tools in contrastive studies using a descriptive approach. This approach consists in presenting and in discussing the characteristics or features of various correlations that will be established. To achieve this objective, this work will be structured as follows: characteristics of corpora used in contrastive linguistics will be presented in the first part; some current trends of methodological operations in the discipline will be discussed under the second part; some useful principles will be identified afterwards; and the last part will consist in establishing and in classifying correlations.

2. Characteristics of corpora used in contrastive linguistics

The characteristics of corpora used in contrastive linguistics can be inferred directly

from the history of the discipline and the theoretical trend in vogue in the field of general linguistics. There is no consensus about the birth of contrastive linguistics. Scholars have divergent views on its date of birth and on its place of birth (see Bayiha's thesis). In fact, some authors posit that CL saw the daylight in the 19th century in Europe with authors like F. von Schlegel, Franz Bopp, Humboldt; whereas others claim that CL has rather come into existence in the 1950s in US with authors like Lado, Selinker, Fries, etc. The historical route of the use of corpora is rather different with that of CL even though the history of the former is intimately linked to that of the latter. It is worth noting that our concern here is not presenting the characteristics of a specific type of corpus, but those of corpora in general within the field of contrastive linguistics. To guarantee a good grasp of it, these characteristics will be presented in two phases.

2.1 Before the 1990s

The characteristics of corpora used in contrastive linguistics, which are presented in this study, are drawn from Hasselgard (2020). As already mentioned, there are two views on the birth of contrastive linguistics. Whatever view is considered, corpora were predominantly monolingual. It means that subcorpora were examined much more in isolation. Furthermore, corpora were obtained from systems in the abstract. In other words, they were not made up of language aspects in context, but of mere set of data. These data were just chunks of languages (sounds, phrases, sentences) without any specific data collection technique or any specific collection guideline. Such corpora were nameless and tools for their collection were not clearly stated.

This characteristic trend is motivated by the tender age of the discipline which is a period during which philosophy on contrast has emerged with authors like Humboldt, Otto Jespersen, Whorf and that has witnessed the waves of transition in theoretical linguistics with major scholars like Lado and Fries according Pan and Tham (2007).

2.2 As from the 1990s

Johansson (2003), Xiao (2010), and Hasselgard (2020) agree that the revival of contrastive linguistics has taken place in the 1990s because of a new impulse in the methodological perspective with the advent of multilingual corpora. Such corpora integrate subcorpora [considered as elements that make up a corpus by Fonkoua and Bayiha (2023)] from different languages, and gathered based on some criteria. It means that there is a link between the components of a corpus. These corpora set a solid basis to the discipline contrarily to what characterized corpora before the 1990s. As from the 1990s, scholars have become deeply interested in theorizing the issue of corpus in contrastive linguistics, that is, by investigating on corpora, conceptualizing phenomena related to corpora, questioning previous views on corpora, coining concepts, criticizing lacks, and establishing principles about corpora. These various works resulted in the coining of several corpora classification (*monolingual vs. multilingual, translation corpus vs. comparable corpus vs. translation comparable corpus, unidirectional corpus vs. bidirectional vs. multidirectional corpus*) and the identification of criteria useful for it (*number of languages, relationship between subcorpora, contrastive direction, themes*). This period is also witnessing a huge debate on the suitable appellation of classes of corpus. For instance, divergences between

scholars can be consulted in Fonkoua and Bayiha (2023: 337-338). Murphy and Riordan (2016) provide an overview of classes of corpus from a thematic perspective that is worth mentioning: general corpora, specialized corpora, parallel corpora, historical corpora, and multimodal corpora.

3. Some current trends of methodological operations in contrastive linguistics

By methodological operations, we mean activities related to corpus preparation in order to carry out a contrastive study on a specific comparative concept. Trends of methodological operations in contrastive linguistics are therefore dependent on one of saussurian dichotomies, which is *langue* vs. *parole*. In this respect, this dichotomy is clarified in Ferdinand De Saussure's Course in General Linguistics (1959: 14):

In separating language from speaking, we are at the same time separating: (1) what is social from what is individual; and (2) what is essential from what is accessory and more or less accidental. Language is not a function of the speaker; it is a product that is passively assimilated by the individual... speaking, on the contrary, is an individual act. It is willful and intellectual.

More precisely, De Saussure, F., (1908-1909:4) defines both concepts in the following ways:

La langue [language] est un ensemble de conventions nécessaires adoptées par le corps social pour permettre l'usage de la faculté du langage chez les individus. La faculté du langage est un fait distinct de la langue mais qui ne peut s'exercer sans elle. Par *la parole* [speaking], on désigne l'acte de l'individu réalisant sa faculté au moyen de la convention sociale qui est la langue. Dans la parole il y a une idée de réalisation de ce qui est permis par la convention sociale.

The English and the French versions of Saussure's explanation of the dichotomy "langue vs. parole" shed more light on what the "langue part" and the "parole part" stand

for. It enables us to deduce that methodological operations for contrasting are carried out at two distinct levels: the 'langue' level and the 'parole' level.

3.1 Revisiting language placement options for contrasting

Languages are bodies of conventions that are not yet used by individuals in a given social context. Pan and Tham (2007) distinguish three placement options when contrasting namely: unidirectional, bidirectional, and multidirectional placements of languages. Before tackling each of them in details, it seems necessary to present a legend that is of a paramount importance for the understanding of what follows.

Legend:

- **Starting language (L1):** it is a position occupied by a language that stands for the comparison starting point. In other words, it is a position occupied by a language from which the comparative work starts.
- **Another language (L2):** it is a position occupied by a language that stands for the comparison completion point. Simply put, it is a position occupied by a language from which the comparative work ends.
- **→ and ←:** indicates the contrastive directions; the **starting language** being at the base of the arrows and the **another language**, at the end of it.
- **0:** indicates any tertium comparationis (see Krzeszowski, 1990)
- **...:** suggests a possible adding of languages in the comparative chain
- **A, B, C :** are languages involved in the contrastive work

3.1.1. Unidirectional placement

This placement refers to the existence of one direction between the compared entities no matter their position with the placement setting. In this case, the contrastive work starts from A to B or from B to A. schematically, it is represented as follows:

$$1) \begin{matrix} A & \rightarrow & B \\ (L1) & & (L2) \end{matrix} \quad \text{or} \quad 2) \begin{matrix} B & \rightarrow & A \\ (L1) & & (L2) \end{matrix}$$

The above schemas portray a contrastive context with two languages. In the first case, A (L1) is considered as a starting language and B (L2), as an ‘another language’; or in the second case, B (L1) stands for the starting language and A (L2), as the ‘another language’. The implementation of the unidirectional placement of languages in a study consists in selecting just one of the alternatives. In this respect, Pan and Tham (2007) suggest to place as a starting language, languages that fulfill the following criteria: linguistic power, documentation, and computerizing. Placing as a starting language, languages that do not obey to these criteria may be risky for the contrastive work based on the unidirectional placement.

Linguistic power does not refer to the history of a language, to the number of speakers, or to the language of the colonizers, but to its international influence. The international influence of language deals with its adequacy of description; documentation meaning the availability of books, research papers on grammar and on other language aspects; and with computerizing, which is a situation whereby language resources are available and accessible to all in the internet.

3.1.2 The bidirectional placement

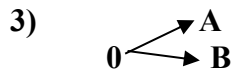
Bidirectional placement indicates the existence of two directions between the compared entities. In this case, the contrastive work has two routes: in the first route, it starts from A (a powerful language) to B (a less powerful language), and in the second route, it rather starts from B to A. Another alternative equally suggests two routes wherein the contrastive activity starts from B (a less powerful language) to A (a powerful language) and from A to B. Schematically, we have:

$$1) \begin{matrix} A & \rightarrow & B & \rightarrow & A \\ (L1) & & (L2) & & (L1) \end{matrix} \quad \text{i.e.} \quad A \rightarrow B \text{ and } B \rightarrow A$$

$$\text{Or} \\ 2) \begin{matrix} B & \rightarrow & A & \rightarrow & B \\ (L1) & & (L2) & & (L1) \end{matrix} \quad \text{i.e.} \quad B \rightarrow A \text{ and } A \rightarrow B$$

It can be noticed that alternatives of the bidirectional placement of languages possess two directions including languages that enjoy one after the other the position of ‘starting language’ and of ‘another language’. Such a placement is appropriate for studies related to back-translation and for studies in favour of endangered or poorly endowed languages. It is therefore advised to select one of the alternatives offered by this placement when carrying out a piece of research and to explore the two routes that make it up. The second case is preferable because it facilitates in attaining the above goals. Simply put, it seems better to place an endangered or a poorly endowed language as a ‘starting language’ and a powerful language as an ‘another language’ in the first route so that a less powerful language should be an ‘another language’ in the last route. Contrarily to powerful languages, less powerful ones are less codified, less documented, less influent internationally, less computerized, and less adequately described.

Another case emerges under the bidirectional placement when considering a contrastive study based on a tertium comparationis and on two languages. Such a case shows that the contrastive work can start from the tertium comparationis to the languages involved in the comparison. Pan and Tham (2007) view this case as *one way* and the previous discussed in above paragraphs as *double way*. This case can be labelled as followed:



This case is named *one way* because of the sense of the contrastive work between the tertium comparationis (0) and the languages A excluding any form of comparative reversibility. This case is part of the bidirectional placement because there exist two contrastive directions, which can be exploited during a study: the first one starting from a tertium comparationis to A; and the last one starting from a tertium comparationis to B.

In order to implement this case in a study, the contrastivist should make sure that the selected tertium comparationis exists in the two linguistic systems involved in the study. By tertium comparationis, we mean any language aspect or topic that first exists, that can be implemented, that is identified within the linguistic system of compared languages and then, that serves as a comparative common ground during the study.

3.1.3. The Multidirectional placement

The multidirectional placement indicates the existence of more than two directions between multiple compared entities. The major difference between this placement and the bidirectional one lies on the number of directions and the number of compared entities, but the comparative principle remains the same. This placement offers two cases according to Pan and Tham (2007): one-to-many and zero-to-many.

The first case called 'one-to-many' suggests that one language is considered as a 'starting language' and other ones are 'another languages' in such a way that the comparative relationship exists between the language, considered as the 'starting language', and 'another languages' individually. Languages considered as 'another languages' have a common

denominator i.e. they are involved in the contrastive work relating each with the 'starting language'. It is schematically represented as follows:

$A \rightarrow B, C, D, E, F, \dots$ i.e. 1) $A \rightarrow B$, 2) $A \rightarrow C$, 3) $A \rightarrow D$, 4) $A \rightarrow E$, 5) $A \rightarrow F$, 6) $A \rightarrow \dots$

We can notice that the behaviour of language A will just be compared to that of languages B, C, D, E, F, ... one after the other. It is good to place powerful languages at the 'starting language' position knowing that power is relative and depends on the languages involved in the contrastive work. For instance, in a set of languages containing English, English will be automatically considered as the 'starting language'. In a set of languages made up for instance of Chinese, French, Spanish, German, Russia, the contrastivist needs to identify which one of them is more powerful before selecting and placing it as the 'starting language'.

The second case called 'zero-to-many', refers to a situation whereby a tertium comparationis and more than two languages are involved in the contrastive work. In this case, all languages are considered as 'another languages' and the tertium comparationis being placed where a 'starting language' must be in the comparative chain. Schematically, we have:

$0 \rightarrow A, B, C, D, E, F, \dots$ i.e. 1) $0 \rightarrow A$, 2) $0 \rightarrow B$, 3) $0 \rightarrow C$, 4) $0 \rightarrow D$, 5) $0 \rightarrow E$, 6) $0 \rightarrow F$, 7) $0 \rightarrow \dots$

In this case, the contrastivist does not need to identify the more powerful language in a set, but he/she rather examines tertium comparationis to select those that are generally universal and those that are specifically applicable to languages involved in the contrastive work. The latter will consist in discovering various manifestations and subtleties of a particular tertium comparationis within each language, one after

the other. There is no comparative link between languages; they rather constitute platforms in which the same tertium comparationis will be applied. The comparative link rather exists between the tertium comparationis and languages.

3.2 Revisiting subcorpora placement possibilities for contrasting

Saussure's dichotomy 'langue vs. parole' well establishes the reason why placement possibilities are separately examined at language level and at corpora level. Furthermore, we compare languages by comparing texts, oral performances or transcribed texts of these languages. Besides, a tertium comparationis is abstract; its realization is only perceptible within the text. In this respect, Fonkoua and Bayiha (2023) thoroughly examine ways in which subcorpora (being constitutive parts of a corpus) can be placed for a contrastive study.

Summarily, Fonkoua and Bayiha (2023) identify three contrastive directions at the corpus level namely: unidirectional, bidirectional and multidirectional. For a good grasp of schemas highlighting various contrastive directions, a small legend is presented:

ST (source Text): is a subcorpus from Language A
 $\rightarrow \leftarrow$: reversible direction

TT (target text): is a subcorpus from language B
 $\rightarrow$: one direction

TT1: one version of target text from a ST

TT2: another version of TT in another language from the same ST

TT1': another version of target text in the same language from the same ST

3.2.1 Unidirectional placement

The unidirectional placement of subcorpora refers to a placement where there exists one contrastive direction between the compared texts: one considered as the source

text and the other, as the target text. In our humble opinion, a text is acknowledged as being source if it is the raw material for a translation activity; whereas, a text is viewed as target if it is a translation product. It is worth to note that the unidirectional placement also includes a type of placement wherein the target text is considered as the starting point for comparison. The unidirectional placement is illustrated below:

$ST \rightarrow TT$ or $TT \rightarrow ST$

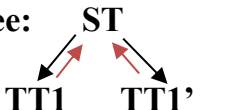
3.2.2 Bidirectional placement

This placement highlights a contrastive reversibility between the compared entities. By contrastive reversibility, we mean that each text, be it source or target, is considered as the starting point for the contrastive work. The bidirectional placement has three degrees:

1st degree: $ST \rightleftarrows TT$ i.e. $ST \rightarrow TT \rightarrow ST$ or $TT \rightarrow ST \rightarrow TT$

The 1st degree of bidirectionality presents a comparative reversibility between a source text and a target text. Considering Naomi Alderman's novel, *Disobedience* originally written in English and translated into *La desobéissance*, the French version. For instance, Both can constitute a good bidirectional placement.

2nd degree:



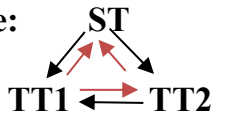
$ST \rightarrow TT1 \rightarrow ST$ or $TT1 \rightarrow ST \rightarrow TT1$

$ST \rightarrow TT1' \rightarrow ST$ or $TT1' \rightarrow ST \rightarrow TT1'$

The 2nd degree of bidirectionality presents two pairs of comparative reversibility: a first one between the source text and one version of the target text in a language; and the second one between the source text and another version of the target text in the same language. Considering comparative setting made up of the New Testament of the Bible written in Latin, translated into English (King James

version), and translated also into English (Amplified version); there is a comparative reversibility between Latin version and King James version; and a comparative reversibility between Latin and Amplified version, still written in English.

3rd degree:

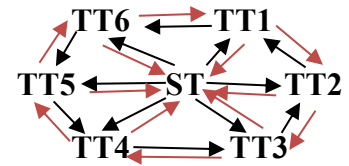


ST → TT1 → ST or TT1 → ST → TT1
 ST → TT2 → ST or TT2 → ST → TT2
 TT1 → TT2 → TT1 or TT2 → TT1 → TT2

The 3rd degree of bidirectionality lies on three pairs of comparative reversibility existing between entities i.e. between the source text and a version of the target text in one language; between the source text and another version of the target text in a different language; and between the two versions of the target text. Let us consider a situation that can illustrate such a placement: the New Testament of the Bible originally written in Latin, translated into French (version Louis Segond), and translated into English (King James Version). Comparative reversibility can therefore be established between the source text in Latin and the first target text in French; between the source text in Latin and the second target text in English; and between the first target text in French and the second target text in English.

3.2.3. Multidirectional placement

The multidirectional placement refers to a placement with multiple contrastive directions between multiple entities. This placement is very difficult to be implement maybe except in carrying out large-scale linguistic projects and in a long period. In such a context, there are a source text and several target texts of different languages having multiple comparative relationships. It can be represented as follows:



This placement is possible with UN texts that have translated versions into UN official languages or with widespread religious texts like the Bible that has been widely translated. This placement equally offers multiple contrastive options. It is worth mentioning that a tertium comparationis cannot be a starting point for comparison in subcorpora placement because it is not a text, and consequently cannot be consider as a source text, but it is rather its realizations that are identified within subcorpora.

To sum up, despite some similarities that exist between language placements and subcorpora placements, it is obvious that they are different. These differences stem from the fact that texts, constituting subcorpora, are produced from languages. In other words, with one language, a wide variety of texts can be produced and it is in a text written in a specific language that the genius of that language can be seized. Tertium comparationis are displayed within texts. In fact, texts are just platforms where we can seize the inner functioning of a language and the display of a given tertium comparationis. In this regard, language placements guide the placements of subcorpora and reversely, subcorpora placements complement that of languages and facilitate research on languages. Appropriate correspondences between these groups of placement are governed by some principles.

4. Some useful principles

Placements are not done at random and appropriate correspondences between language placements and subcorpora are not established haphazardly. They have to follow principles that are guidelines for a research

study. Pan and Tham (2007) discuss issues related to methodological principles that will be presented in the table below:

Table 1. Methodological principles according to Pan and Tham (2007)

Principles	Definition	Nature	Examples
Principles of background	Those decisive factors behind methodology (Pan and Tham 2007:234)	Compulsory	-Ontology-decisive -Teleology-decisive -Language-perspective-decisive
Principles of selection	Principles that allow researchers some freedom of choice before they set about their work. (Pan and Tham 2007: 237)	Subjective	-Selection for language view and linguistic theory -Selection for strata of research -Selection for levels of research -Selection for direction of contrasting -Diversity in research methods -Selection of research models
Principles of homogeneity	Are to a certain extent restrictions on the principles of selection (Pan and Tham 2007: 239)	Restrictive	-Consistency in theory -Homogeneity of linguistic data -Correspondence in domains -Strata by gradation

Details on the above principles are found in Pan and Tham (2007). With this table, we realize that there are three groups of principles sustaining the methodology of contrasts in linguistics namely: principles of background, principles of selection, and principles of homogeneity.

The principles of background are not directly seen in a study, but are compulsory basic guidelines for a contrastive work. Moreover, they are rather decisions made before the start of the study in order to give an orientation to it. It means that ontology, teleology, and the language-perspective can normally be inferred from each contrastive study.

The principles of selection are subjective in nature, that is, they thoroughly depend on the researcher. These principles stipulate that the contrastivist must make choices at

various levels of the study. This group of principles seems to go against to the main research principle that is objectivity because the contrastivist may intend to select at random without any methodological justifications, putting the study into the background. Therefore, the principles of homogeneity intervene to compensate for the shortcomings of the principles of selection.

The principles of homogeneity are considered as restrictions on the principles of selection. They restore the principle of objectivity in a contrastive study. Even if the contrastivist has to select at various levels of the work, his/her selections need to be homogeneous. It means that there must be an appropriate correspondence link between different selection levels or within a specific

selection set as there are listed in the table above.

Within the framework of this study, the selection for the direction of contrasting, which falls under the principles of selection, can be done both at the language level and at the subcorpora level. For the task to be devoid of subjectivity, the correspondence in domains, which falls under the principles of homogeneity, will intervene to assure the establishment of an appropriate correspondence link between potentially selected directions of contrasting at language level and those at subcorpora level. This point is the core of the next part of this study. Viewed in that sense, we realize that the principles of homogeneity allow a contrastive study to meet general research principles like objectivity and consistency.

5. Correlations

This part aims at facilitating the task of contrastivists in enacting the principles of homogeneity while selecting directions of contrasting. We intend to present appropriate correspondence links (correlations) between language placements and subcorpora placements for contrasting. Before establishing correlations, the census of various placements at language level and at corpus level based on separate examination that we have carried out above:

Table 2. Census of various placements

Language placements	Subcorpora placements
1- Linguistic unidirectionality	1- Corpus unidirectionality
2- Linguistic bidirectionality type 1 (double-way)	2- Corpus bidirectionality 1 st degree
3- Linguistic bidirectionality type 2 (one-way)	3- Corpus bidirectionality 2 nd degree
4- Linguistic multidirectionality type 1 (one-to-many)	4- Corpus bidirectionality 3 rd degree

5- Linguistic multidirectionality type 2 (zero-to-many)

5- Corpus multidirectionality

According to the above table, there are five possible placements at language level as well as at corpus level. All these placements are identified with the help of a common nomenclature (*unidirectionality*, *bidirectionality*, and *multidirectionality*). Apart from these similarities, some differences can be brought out: while linguistic bidirectionality provides two types of placements (*double-way* and *one-way*), corpus bidirectionality rather offers three degrees of placement (1st, 2nd, and 3rd). Besides, linguistic multidirectionality has two types of placements (*one-to-many* and *zero-to-many*) whereas corpus multidirectionality possesses just one placement.

The consideration of characteristics of each placement suggests the following correlations:

- Linguistic unidirectionality ↔ Corpus unidirectionality
- Linguistic bidirectionality type 1 ↔ Corpus bidirectionality 1st degree
- Linguistic bidirectionality type 2 ↔ Corpus bidirectionality 1st degree
- Linguistic multidirectionality type 1 ↔ Corpus bidirectionality 2nd degree
- Linguistic multidirectionality type 1 ↔ Corpus bidirectionality 3rd degree
- Linguistic multidirectionality type 1 ↔ Corpus multidirectionality
- Linguistic multidirectionality type 2 ↔ Corpus multidirectionality

Out of the five placements on both sides, seven correlations have been established because some language placements have more than one correlation. For instance,

linguistic multidirectionality type 1 has three correlations. In next paragraphs, these correlations will be discussed, and they will be classified as being simple or complex.

5.1 Simple correlations

By simple correlations, we mean correlations whereby correspondence links are easily established. Simply put, when correspondence links are obviously established between entities of the language placement chain and those of the subcorpora placement chain, such a correlation is simple. Among the seven correlations that have been listed above, three of them are simple such as: Linguistic unidirectionality ↔ Corpus

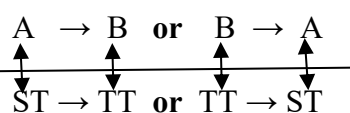
Language level

Subcorpora level

unidirectionality, Linguistic bidirectionality type 1 ↔ Corpus bidirectionality 1st degree, and Linguistic multidirectionality type 1 ↔ Corpus multidirectionality.

5.1.1 Linguistic unidirectionality ↔ Corpus unidirectionality

This correlation accounts for a contrastive setting whereby the contrastive direction is the same at language level and at subcorpora level; and there is a one-to-one correspondence between the placement of languages involved in the contrastive work and the placement of subcorpora; each subcorpus being a sample of a language.



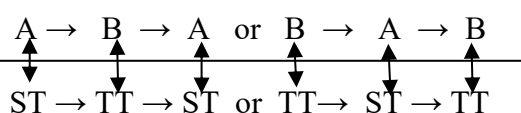
realities of all contrastive placement alternatives both at language and subcorpora levels.

5.1.2 Linguistic bidirectionality type 1 ↔ Corpus bidirectionality 1st degree

This correlation describes the sameness of the contrastive direction both at language and at corpus level; and the correspondence of a double-way direction between two languages in comparison and their respective subcorpus.

Language level

Subcorpora level

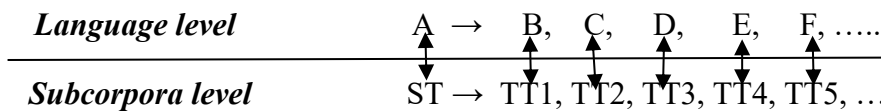


The above correlation can be implemented in a study based on two distinct languages. The most difficult tasks are to find representative subcorpora for languages in comparison; to watch over the homogeneity between the methodological setting and analytical level; and to select one of languages as the starting language. The advantage of this correlation lies on the analytical precision of similarities and differences in a particular contrastive direction. As disadvantage, it scarcely allows researchers to seize comparative

This correlation is also implemented in a study based on two distinct languages. In this case, the challenge is almost the same with the first correlation, i.e. to find representative subcorpora for languages in comparison; to watch over the correspondence link between the methodological setting and the analytical level while carrying out a piece of research;

and to select one of the languages involved in the comparison as the starting language. As advantages, it offers two compulsory contrastive directions that are conducive to the discovery of all contrastive realities between the two selected languages. The placement imposes each of them to be the starting language at one point in time and the

‘another language’ at another. Moreover, the contrastive reversibility $\dots \rightarrow B \rightarrow A$ or $\dots \rightarrow A \rightarrow B$ is no more done with a new subcorpus for A in the first case and B in the last case, but with the initial selected subcorpus. The disadvantageous part of this correlation is the further contrastive work, imposed by the comparative reversibility, suggested by the placement. Methodological realities at this point are very delicate because



In this correlation, there is an indication of the contrastive direction at the language level and which is transposed at the subcorpora level; and there is an indication of the correspondence between entities of the two levels. So, the implementation of such a correlation can be done in a study based on more than two languages. Apart from finding representative subcorpora, the challenge is all about matching appropriately the placement at language level and at subcorpora level; carrying out the contrastive work effectively for each pair of languages; and keeping the correspondence link between the methodological setting and the analytical level for each pair of languages involved in the study. As advantages, it allows to check the renderings adequacy of a same source text into many languages. It equally makes possible the comparison between the source language and each language into which texts have been translated or between translated texts. As weak points, implementing this correlation is time constraint and requires multiples studies.

5.2 Complex correlations

Unlike simple correlations, complex correlations refer to those wherein

it is not a research per se, but a kind of checking process in order to balance findings.

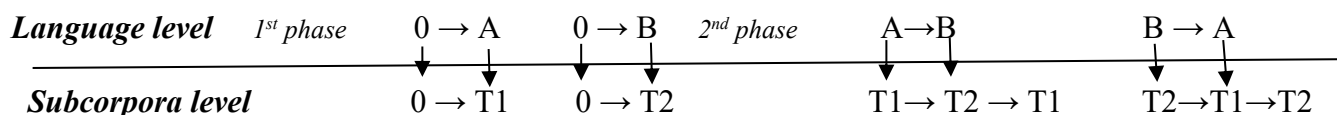
5.1.3 Linguistic multidirectionality type 1 ↔ Corpus multidirectionality

This correlation denotes a context of a multiple-way direction starting from one item to many; and a correspondence between the disposal of languages and of subcorpora since TT1, TT2, TT3, TT4, TT5 are translations into distinct languages of ST.

correspondence links are difficultly established between entities at the language level and at the subcorpora level. As complex correlations, we propose the following: Linguistic Bidirectionality type 2 ↔ Corpus Bidirectionality 1st degree, Linguistic Multidirectionality type 1 ↔ Corpus Bidirectionality 2nd degree, Linguistic Multidirectionality type 1 ↔ Corpus Bidirectionality 3rd degree, and Linguistic Multidirectionality type 2 ↔ Corpus Multidirectionality. Each one of them will be discussed in next paragraphs.

5.2.1 Linguistic Bidirectionality type 2 ↔ Corpus Bidirectionality 1st degree

This correlation is part of complex ones because correspondence links are hard to seize. It is worth to mention that horizontal arrows in the schema below indicate the comparison relations meanwhile vertical arrows shows correspondence links between entities of two levels. The overview of this correlation is below:



The above correlation is implemented in two phases with two distinct languages and a tertium comparationis. There exist two relationships between entities at each level: a realization relationship between the tertium comparationis and languages; and a non-translation relationship between languages or between their respective subcorpora. This type of relationship highlights the way a particular tertium comparationis behaves in selected languages meanwhile the second type shows that languages or their subcorpora are comparable i.e., they are selected based on a common criterion.

the first phase, there are correspondences between the tertium comparationis related to languages and the same tertium comparationis related to subcorpora. The difference between the tertium comparationis at various levels is that it is placed in an abstract environment at the language level whereas it is examined in a more concrete environment at the subcorpora level.

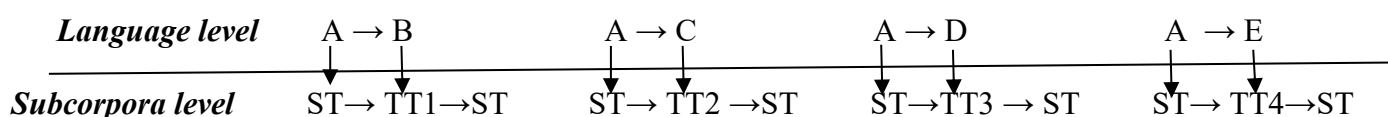
In the second phase, correspondence links are rather established between languages in comparison and subcorpora in such a way that the contrastive work, carried out with subcorpora, is done in two ways following the selected option i.e., $T1 \rightarrow T2$ and $T2 \rightarrow T1$ or

$T2 \rightarrow T1$ and $T1 \rightarrow T2$. Furthermore, the comparison is not done in terms of checking translation renderings but in terms of displaying ways of the tertium comparationis in the two linguistic system.

As advantages, this correlation allows to get the display of a tertium comparationis in two linguistic system in isolation and to compare the ways a tertium comparationis is displayed. The comparative reversibility in this correlation allows to obtain a 'back contrast' that results from checking an initial contrastive work between two languages by reversing the comparative direction between the subcorpora. Nevertheless, it shows some weaknesses: the imposition of stages during the implementation of this correlation may make the work tough and weaken the analytical presentation.

5.2.2 Linguistic Multidirectionality type 1 ↔ Corpus Bidirectionality 2nd degree

This correlation is based on more than two languages, one language being a 'starting language' and the others being 'another languages'. It is classified under complex correlations because there is a comparative reversibility at the subcorpora level in each stage of the contrastive work as shown below:



Although correspondence links are clearly established between entities at language level and at subcorpora level, the comparison relationship between

subcorpora suggests a comparative reversibility at each stage of the contrastive work. In other words, the peculiarity of this correlation is that apart from the

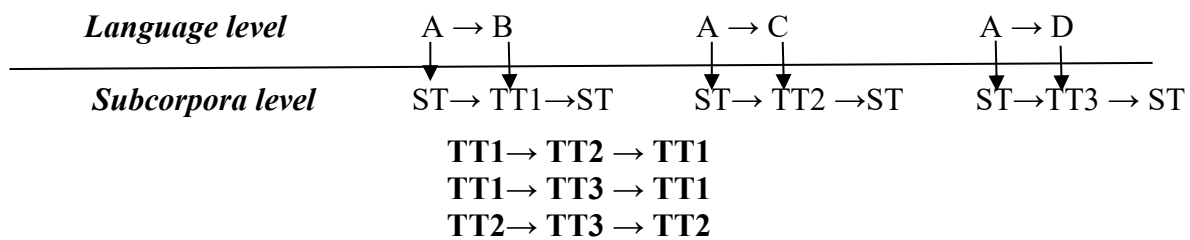
correspondence links, there is a reciprocal contrastive direction between various subcorpora at each comparative set in the whole work.

As strong points, the above correlation favours a detailed examination of linguistic realities of one language with regard to many others; and besides, it widens the scope of a contrastive study in terms of the number of languages. As weak points, it may be hard to implement in a single study because of the existence of more than two comparative sets. The denseness of such a study may give room to cacophony in terms of data presentation and analysis; moreover, the depth of the overall work may be minimized

due to the management of a huge amount of information.

5.2.3 Linguistic Multidirectionality type 1 ↔ Corpus Bidirectionality 3rd degree

This correlation is also part of complex because its features match much more with the characteristics of the latter. It can be implemented in a study of more than two languages thanks to their respective subcorpora in such a way that there two varieties of links: correspondence links existing between entities at the language level and at the subcorpora level, comparative links existing between languages and between subcorpora. This correlation is schematically represented as shown below:



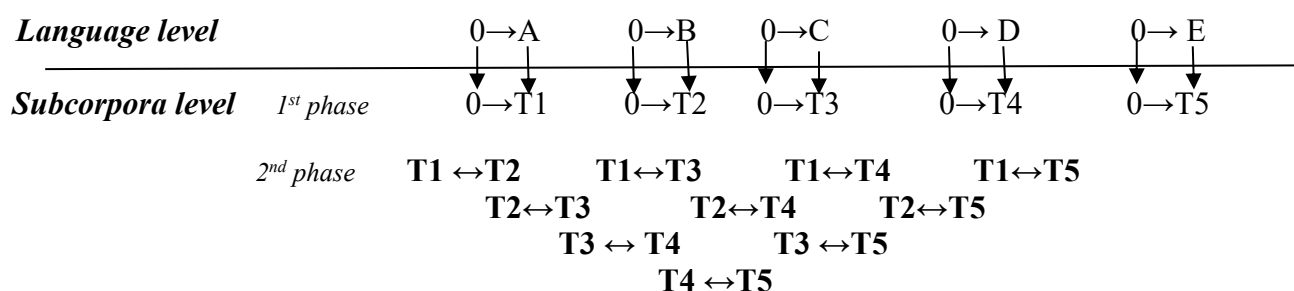
The complexity of this correlation stems from the existence of correspondence links between some entities at the language level and at the subcorpora level; the presence of comparative links between languages, subcorpora with a comparative reversibility; and the existence of comparison relationship between various subcorpora of 'another languages' (TT1, TT2, TT3) with a comparative reversibility.

For instance, with four languages (A, B, C, D), a total of six comparative stations can be constituted with three being directly linked to the placement of languages and the three others being isolated, i.e. not directly linked to the placement of languages, but to the number of subcorpora. This correlation shares the same advantages and disadvantages with the previous correlation,

with the sole difference that it offers more comparative stations.

5.2.4 Linguistic Multidirectionality type 2 ↔ Corpus Multidirectionality

This correlation is one of the most complex because its implementation requires more than two languages, the overall comparative work is divided into comparative moments in the two phases, and because of the presence of two contrastive directions namely: unidirectional in the first phase and bidirectional in the second phase. The above-mentioned can be viewed in the following schema:



This correlation shares common features with the first complex correlation (*Linguistic bidirectionality type 2 ↔ corpus bidirectionality 1st degree*). The difference between them is that the comparative task moves from *tertium comparationis* → *subcorpoora* to *subcorpora* → *subcorpora*, each subcorpus enjoying the position of the starting point of the contrastive work. According to the above schema, there are perfect correspondences between entities at language and those at the subcorpora level. In the first stage, the comparative work includes the *tertium comparationis* and various subcorpora from languages. In a practical way, the task is done progressively and consists in identifying realization forms of the *tertium comparationis* within each subcorpus. In the second stage, the contrastive task includes subcorpora and consists in comparing reversibly the realization forms of the *tertium comparationis* in one specific subcorpus with another. It is equally done progressively depending on available pairs of subcorpora. For instance, with five languages, there will be five comparative moments in the first phase of the overall study, and ten reversible comparative pairs in the second phase meaning that each subcorpus enjoys the starting point of the work.

6. Synthesis

An overall consideration of all correlation types suggest some remarkable facts leading

to the identification of contexts where they are susceptible to be implemented, and their classification as being a reduced version of some types or an extended version of others.

6.1 Remarkable facts on various correlations

As far as simple correlations are concerned, they all share common features. For instance, their implementation is relatively simple as the name suggests; they are operationalized in translation contexts by opposition to comparable contexts and comparable translation contexts. In fact, they account mostly for translation corpora.

As for complex correlations, they do not share common features, except their complexity. So, they are implemented in translation contexts, non-translation contexts, and hybrid contexts.

To sum up, these correlations are not concerned with monolingual frames, but with multilingual contexts because the least of the types is implemented with the aid of two different languages. They are linked one way or the other either in terms of the number of languages, or the reducedness and the extendedness, or in terms of their characteristics, and or the nature of correlations to which they belong. All correlation types pursue a common objective that is to facilitate the contrastive work within the field of contrastive linguistics.

6.2 Translation contexts, non-translation contexts, and mixed contexts

This trilogy sheds light on contexts where various correlations can be implemented. These contexts are three in number: translation contexts, non-translation contexts, and mixed contexts. They are coined based on the three types of corpora in contrastive linguistics namely: translation or parallel corpora, comparable corpora, and parallel comparable corpora (see Fonkoua and Bayiha, 2023).

Translation contexts clearly allow the identification of the source language or text and the target language or text. Contrastive studies, based on such contexts, deal with the comparison of at least two languages based on at least two texts (source and target). Among the correlations presented in this study, some of them are implemented only within translation contexts:

- Among simple correlations, there are *linguistic unidirectionality* ↔ *corpus unidirectionality*, *linguistic bidirectionality type 1* ↔ *corpus bidirectionality 1st degree*, and *linguistic multidirectionality type 1* ↔ *corpus multidirectionality*
- Among complex correlations, there is *linguistic multidirectionality type 1* ↔ *corpus bidirectionality 2nd degree*

Non-translation contexts allow the identification of languages selected thanks to a criterion that may be a comparative concept or a tertium comparationis. Contrastive studies based on these may examine the way a comparative concept is realized within languages through particular texts. They may extend to compare various realizations of a comparative concept in at least two different texts of different languages. Among the correlations

discussed in this study, some of them are exclusively operationalized within non-translation contexts:

- There is no correlation among simple correlations
- Among complex correlations, there are *linguistic bidirectionality type 2* ↔ *corpus bidirectionality 1st degree*, and *linguistic multidirectionality type 2* ↔ *corpus multidirectionality*.

Mixed contexts include the realities of the above types of contexts. There is just one type among complex correlations that is *linguistic multidirectionality* ↔ *corpus bidirectionality 3rd degree*

6.3 Reduced and extended versions

This dichotomy accounts for relationships that exist between various correlations that are classified under simple and complex correlations.

All simple correlations can be placed in a continuum starting from the most reduced version to the most extended version. Clearly, *linguistic multidirectionality type 1* ↔ *corpus multidirectionality* is more extended than *linguistic bidirectionality type 1* ↔ *corpus bidirectionality 1st degree* because the former is obtained by adding of more entities at the language level entailing the adding of their counterparts at the subcorpora level. The latter correlation is more extended than *linguistic unidirectionality* ↔ *corpus unidirectionality* because of the same reason. Reversely, *linguistic unidirectionality* ↔ *corpus unidirectionality* is the reduced version of *linguistic bidirectionality type 1* ↔ *corpus bidirectionality 1st degree* because the former correlation is obtained by subtracting some entities of the latter correlation. *linguistic bidirectionality type 1* ↔ *corpus bidirectionality 1st degree* is the reduced version of *multidirectionality type 1*

↔ *corpus multidirectionality* because of the same reason.

Relationships can also be established between the constituents of complex correlations. In terms of reducedness, a panoramic view of various types of complex correlations suggests that *linguistic bidirectionality type 2* ↔ *corpus bidirectionality 1st degree* is the reduced version of *linguistic multidirectionality type 2* ↔ *corpus multidirectionality*. The reason of such a link is that the former is obtained by subtracting the stage of the contrastive work consisting in a reversible comparison of subcorpora. Moreover, that same link exists between *linguistic multidirectionality type 1* ↔ *corpus bidirectionality 2nd degree* and *linguistic multidirectionality type 1* ↔ *corpus bidirectionality 3rd degree*. In terms of the extendedness, *linguistic multidirectionality type 2* ↔ *corpus multidirectionality* can be considered as the extended version of *linguistic bidirectionality type 2* ↔ *corpus bidirectionality 1st degree* because both of them have the same matrix on which the reversible comparison of subcorpora is added in the former correlation. The same reality exists between *linguistic multidirectionality type 1* ↔ *corpus bidirectionality 3rd degree* and *linguistic multidirectionality type 1* ↔ *corpus bidirectionality 2nd degree*.

7. Conclusion

This study focused on identifying and discussing various types of correlations that are susceptible to be operationalized within the field of contrastive linguistics. We advocated that the selection for the contrastive direction at language level has to be consistent with that of the contrastive

direction of subcorpora according to the principles of homogeneity. This issue brings forth the necessity of establishing correlations between alternative placements of languages for comparison and placement possibilities of subcorpora. Seven correlations have been identified and discussed thanks to the descriptive approach. Their implementation may bring new methodological insights and may positively impinge on the quality of findings.

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